PRACTICAL 7 : Hall Effect Virtual LAB.

Source: https://mpv-au.vlabs.ac.in/modern-physics/Hall_Effect_Experiment/experiment.html

Output:

Hall Coefficient : 0.0194

Carrier Concentration : 3.22165e+20

Calculations:

1. Hall Coefficient Formula

The Hall coefficient R_H is defined as:

$$R_H = \frac{E_H}{J \cdot B}$$

where:

- E_H = Hall electric field (V/m)
- J = current density (A/m^2)
- B = applied magnetic field (Tesla)

It can also be expressed in terms of carrier concentration:

$$R_H = \frac{1}{nq}$$

where:

- n = carrier concentration (m^{-3})
- q = charge of carrier (Coulomb, for electrons $q = -1.6 \times 10^{-19}\,C$)

- Carrier concentration: $n = 3.22165 \times 10^{20}\,\mathrm{m}^{-3}$
- Charge of electron: $q = 1.6 \times 10^{-19}\,C$

$$R_H = \frac{1}{nq} = \frac{1}{(3.22165 \times 10^{20})(1.6 \times 10^{-19})}$$

$$R_H = \frac{1}{5.15464 \times 10^1} \approx 0.0194\,\mathrm{m^3/C}$$